But how does Bitcoin actually work? Watch the following video to understand better.

So far, we've learned about the history of money and the revolutionary idea of *blockchain* technology, and we've explored the basics of ***Bitcoin*** - the world's first decentralized digital currency. But how does ***Bitcoin*** prevent fraud and ensure that people can't spend the same money twice? That's where the magic of UTXOs, public key cryptography, hashing, scripting, and the mempool come into play. Just like a fingerprint ensures that nobody else can use your identity, hashes in ***Bitcoin*** ensure that transactions can't be altered. Scripts are like the rules for a game, making sure transactions follow specific conditions. UTXOs are like the building blocks of a puzzle, keeping track of all the money in your virtual wallet. And the mempool acts like a holding area, making sure all transactions are verified before being added to the *blockchain*. So, let's dive in and discover how ***Bitcoin*** solves the double-spending problem and ensures the integrity of every transaction on its network.

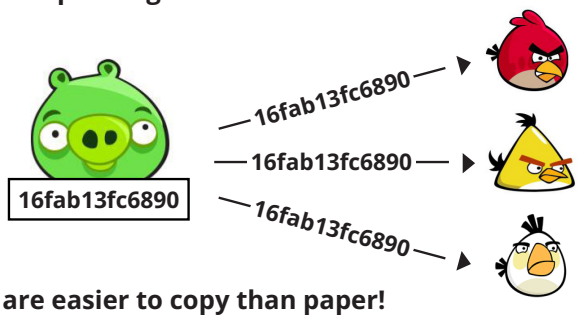
7.0 Putting the Double-Spend Issue to Rest: Understanding Bitcoin's Solution

Remind me, what is the "double spending problem"?

Bitcoin transactions, as we've experienced, are represented by a series of random numbers and letters that can be viewed on any device with internet access.

Additionally, much of the information related to these transactions is typically communicated using a numerical notation system known as **hexadecimal numbers**.

Double spending ...



Bits are easier to copy than paper!

This means that it is common to see strings of 64-character hexadecimal numbers consisting of letters (A-F) and numbers (0-9), such as

f4184fc596403b9d638783cf57adfe4c75c605f6356fbc91338530e9831e9e16

So how do we actually prevent someone from copying and pasting their bitcoin and spending it multiple times, like they would with an email or photo?